

ZHANGJIAKOU, China

WMO: 544010

Lat: 40.7694N

Lon: 114.9194E

Elev: 774

StdP: 92.36

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.2	-15.3	-29.8	0.3	-11.6	-27.7	0.3	-10.7	8.5	-6.8	7.5	-8.2	3.1	340	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	33.1	18.9	31.5	18.6	30.1	18.5	22.8	28.5	22.0	27.5	21.2	26.5	2.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	17.4	25.9	20.3	16.4	25.4	19.4	15.5	24.7	72.1	28.6	68.9	27.6	65.6	26.4	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	7.8	6.5	5.6	DB	-20.0	35.6	2.7	2.0	-21.9	37.0	-23.5	38.2	-25.0	39.3	-26.9	40.7
(5)				WB	-20.7	24.2	2.5	1.1	-22.5	25.1	-24.0	25.7	-25.3	26.4	-27.1	27.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-7.7	-3.7	3.5	11.7	18.0	22.4	24.6	23.0	17.8	10.2	1.3	-5.9	(6)
(7)		DBStd	11.96	3.81	4.55	4.82	4.59	3.86	3.06	2.48	2.40	3.38	3.93	4.65	3.88	(7)
(8)		HDD10.0	1974	550	384	207	34	1	0	0	0	44	262	492		(8)
(9)		HDD18.3	3708	808	617	460	204	52	4	1	48	252	511	750		(9)
(10)		CDD10.0	1857	0	0	6	84	251	372	453	404	234	52	1	0	(10)
(11)		CDD18.3	548	0	0	0	4	43	127	195	147	32	1	0	0	(11)
(12)		CDH23.3	4708	0	0	1	62	472	1136	1703	1080	247	7	0	0	(12)
(13)		CDH26.7	1578	0	0	0	11	143	414	640	319	51	0	0	0	(13)
(14)	Wind	WSAvg	2.4	2.6	2.6	2.7	2.8	2.8	2.3	2.1	2.1	2.1	2.2	2.3	2.5	(14)
(15)	Precipitation	PrecAvg	399	2	4	9	15	30	55	107	106	44	18	5	1	(15)
(16)		PrecMax	591	11	26	32	71	115	107	312	271	109	68	21	6	(16)
(17)		PrecMin	227	0	0	0	0	4	22	15	34	11	1	0	0	(17)
(18)		PrecStd	94	3	5	7	14	21	19	59	56	21	15	5	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	10.5	20.8	28.3	33.0	35.1	35.4	33.6	30.5	24.1	15.4	5.7	(19)
(20)		MCWB	-1.2	2.2	7.9	12.8	15.5	17.6	20.3	19.9	17.4	13.7	6.5	-0.4		(20)
(21)		2%	DB	2.1	7.9	16.9	25.0	30.2	32.6	33.4	31.2	28.1	21.2	12.6	3.5	(21)
(22)		MCWB	-2.9	0.8	6.3	11.3	14.6	17.5	20.2	19.7	16.8	11.3	5.1	-1.6		(22)
(23)		5%	DB	0.2	5.8	14.3	22.6	28.0	30.8	31.7	29.8	26.1	19.1	10.3	1.7	(23)
(24)		MCWB	-4.1	-0.4	4.9	10.2	13.7	17.0	20.0	19.5	15.9	10.4	3.8	-2.8		(24)
(25)		10%	DB	-1.5	3.8	11.7	20.2	25.9	29.1	30.2	28.5	24.3	17.1	8.2	0.0	(25)
(26)		MCWB	-5.4	-1.5	3.6	9.1	12.9	16.8	20.0	19.3	15.0	9.7	2.8	-4.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.9	3.3	8.9	14.7	18.3	21.2	24.5	23.5	20.4	15.0	7.6	0.2	(27)
(28)		MCDWB	4.4	8.5	18.7	22.9	27.6	28.9	30.4	29.3	26.8	21.0	12.0	4.8		(28)
(29)		2%	WB	-2.6	1.2	6.9	12.7	16.7	20.3	23.2	22.6	18.7	13.3	5.9	-1.4	(29)
(30)		MCDWB	1.8	6.8	15.8	21.9	25.4	27.4	29.3	27.7	23.9	18.5	11.5	3.2		(30)
(31)		5%	WB	-4.1	-0.1	5.4	11.3	15.6	19.4	22.5	21.8	17.5	12.0	4.6	-2.6	(31)
(32)		MCDWB	0.0	5.5	13.3	20.3	24.3	26.3	28.3	26.8	23.5	17.2	9.7	1.4		(32)
(33)		10%	WB	-5.3	-1.4	3.9	10.0	14.4	18.6	21.7	20.9	16.5	10.6	3.1	-3.8	(33)
(34)		MCDWB	-1.7	3.6	11.3	18.2	23.0	25.5	27.2	25.7	22.3	15.5	8.0	-0.2		(34)
(35)	Mean Daily Temperature Range	MDBR	9.2	10.2	11.2	11.9	12.0	11.0	9.6	9.7	10.7	10.3	9.1	8.7		(35)
(36)		5% DB	MCDBR	10.9	12.3	14.7	15.7	15.5	14.1	12.4	11.7	13.2	13.0	11.5	10.5	(36)
(37)		MCWBR	7.8	7.9	8.1	7.3	6.1	4.8	4.3	4.5	5.4	6.6	6.9	7.2		(37)
(38)		5% WB	MCDBR	10.6	11.6	13.8	13.2	12.2	10.9	9.3	9.0	10.3	10.7	10.6	10.1	(38)
(39)		MCWBR	7.6	7.6	7.8	6.6	5.4	4.2	4.0	4.1	5.0	6.0	6.6	7.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.390	0.419	0.483	0.537	0.628	0.598	0.583	0.508	0.445	0.354	0.325		(40)
(41)		taud	2.227	2.029	2.000	1.910	1.808	1.679	1.789	1.799	1.908	1.992	2.194	2.233		(41)
(42)		Ebn at Noon	821	793	819	793	759	690	707	701	724	722	764	775		(42)
(43)		Edh at Noon	96	138	159	186	211	241	214	206	173	141	99	89		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.60	3.50	4.79	5.69	6.36	5.85	5.61	5.34	4.57	3.63	2.67	2.25		(44)
(45)		RadStd	0.17	0.22	0.36	0.36	0.43	0.47	0.30	0.34	0.26	0.23	0.27	0.17		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	-1.35	N/A	N/A	N/A	-0.41	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.37	N/A	N/A	N/A	N/A

Nomenclature: See separate page